Frank Schlehofer

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EDUCATION

Purdue University, West Lafayette, IN — December 2026
Bachelor of Science in Computer Science — **GPA: 3.98/4.00**

Relevant Coursework: Analysis of Algorithms, Artificial Intelligence, Data Structures and Algorithms, Systems Programming, Computer Architecture, Relational Databases, Object-Oriented Programming, Discrete Math, C Programming, Statistics, Probability, Calculus II & III, Linear Algebra

PROJECTS

SlayterHiL Hardware-in-the-Loop Testing and Flight Simulation | C++ — September 2025 - Present

- Accelerated development and validation of a custom quadcopter flight controller by contributing to a HIL simulation platform, enabling rapid and safe automated firmware testing.
- Led flight simulation team, engineered physics engine entirely from scratch, implemented a rigid-body dynamics model based on Newton-Euler equations to accurately simulate 6-DoF motion.
- Programmed realistic propulsion model to translate flight controller's motor commands into precise forces and torques required for flight which form the deterministic core of the simulation environment.

Frankie's Fridge | Node, React, Express, REST API, JavaScript — March 2025 – July 2025

- Developed full-stack web application, using Node.js and Express for backend server with RESTful API endpoints and React.js styled with CSS for responsive and intuitive fridge UI.
- Integrated PostgreSQL for persistent data storage with scalable, modular code architecture.
- Practiced full agile methodology with weekly sprints and version control via GitHub.
- Connected with both Google Cloud Vision and Spoonacular API for additional features.

Hybrid Movie Recommendation System | Python, PyTorch, Pandas, Scikit-Learn, Matplotlib — June 2025

- Developed content based and collaborative filtering model in PyTorch to predict movie ratings.
- Engineered a complete ML pipeline, including exploratory data analysis, data preprocessing, and batch loading using PyTorch Dataset and DataLoader.
- Systematically diagnosed and solved model performance issues, including overfitting and unconstrained outputs, improving the validation RMSE from over 5.0 to 1.05.
- Conducted hyperparameter tuning experiments for learning rate and model complexity to optimize performance.

EXPERIENCE

Undergraduate Teaching Assistant, Purdue University — Jan 2025 – Present
Foundations of C, Object-Oriented Programming (Java)

- Instructed weekly lab/office hour sessions to reinforce fundamental C and Java programming concepts.
- Debugged and graded work for 700+ students with individualized feedback to improve coding practices.
- Assisted students with debugging, memory management, and algorithm design.

SKILLS

Programming: Java, C, C++, Python, JavaScript, HTML, CSS, x86 Assembly, Yacc, Lex, Swift
Frameworks/Tools: React, Node, Express, PostgreSQL, Git, OpenCV, TensorFlow, PyTorch, Pandas, Scikit-Learn
Technologies: AWS, Google Cloud Vision API, LaTeX, Supabase, Render, Vercel, Netlify